AI-Based Extraction of Structured Lifespan Data from Scientific Literature

Rafikova E[1,2], Orden PD[2], Nemirovich-Danchenko N[2], Duhart JM[1], Buslje CM[1], Ceriani MF[1]

[1] Fundación Instituto Leloir, Buenos Aires, Argentina [2] Open Longevity, Sherman Oaks, CA, USA

Open Genes

Manually curated database of genes associated with aging and lifespan

Rafikova et al., NAR 2024

1.700 articles

10.000 records

51 fields per lifespan experiment

Manual curation cannot keep pace with the literature

- **Intervention & treatment** — method · genotype · tissue · start · end · stage
- **Animals & housing** — strain · sex · cohort size · diet · temperature · density
- **Lifespan values** — min · mean · median · max
- **Lifespan change** — % change · significance · test
- **Target gene expression** — % change · tissue
- **Aging-related processes** — improved · deteriorated

1. Extraction pipeline

Follows Open Genes curation guidelines and controlled vocabularies; code runs a fixed sequence of steps. Output: one row per field, auditable by the curator.

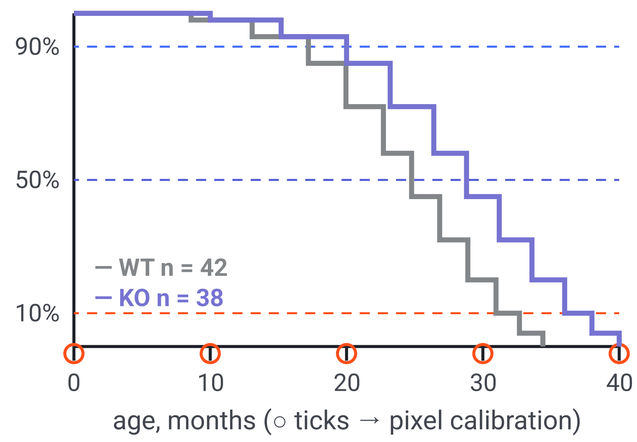
AI model · Python code · Input / output

Article PDF + supplementary — supplementary as docx · doc · pdf	Input
Preparation — PDF text, OCR of figures, supplementary reader	Python
Experiment enumeration — experiment–control pairs by inclusion rules + curve figure	Gemini 3.6 Flash
Paper-level facts — organism, diet, housing, statistical method	Gemini 3.6 Flash
Survival-curve reading — lifespan at 90 / 50 / 10% survival, cohort size, sex	GPT-6 Sol + code
51 fields per experiment — value, evidence, confidence; unanswered fields re-queried	Gemini 3.6 Flash
Consolidation — unsupported values, % change recalculation, rule violations	Python
Record per experiment — + review spreadsheet for the curator	Output

Field	Value	Evidence (quote)	Conf.	Verified
organismLine	C57BL/6	"…derived from the same C57BL/6…"	high	✓
controlCohortSize	56	"WT (n = 56)"	high	✓
lifespanMaxControl	973	curve reading (spread ±4)	high	✓

2. Reading survival curves

GPT-6 Sol writes and runs Python on the figure image; two independent readings per figure.



1. **Locate** — figure page cited by enumeration
2. **Calibrate** — axes from tick-label pixels
3. **Identify** — each group's curve from legend
4. **Measure** — time at 90 / 50 / 10% survival
5. **Read** — time unit, cohort size, sex
6. **Check** — two readings ≤8% apart → accepted

97% of values within 5% of reference
error at 10% survival: 3.2% with code execution vs ~23% without

3. Subpipelines

Supplementary reader

Lifespan values computed by code from per-animal tables; methods passed to the model

Quote verification

each quote searched in the PDF, supplementary text and figure OCR; unsupported values rejected

Guideline rules in code

% change recalculated from absolute values; source priority: text > per-animal data > curve

4. Technology

AI models

Gemini 3.6 Flash (Google API) · GPT-6 Sol with Code Interpreter (OpenAI API)

Documents & OCR

pdfplumber · python-docx · antiword · Tesseract 5.5 OCR · openpyxl

Code

Python 3.10, in-house orchestrator · Pydantic schemas · pytest

5. Results

Run of 26 Sep 2026: 20 mouse-based papers; compared with Open Genes records across 49 fields

43 / 46 eligible experiments identified (93.5%)

86.1% correct fields (1,815 / 2,107)

1.5 h processing time, 20 papers down from 5.5 h

$4.70 API cost, 20 papers down from $14.10

3 missed experiments: lifespan data is in supplementary

Accuracy per paper: 73–98%

errors from supplementary-only data

Two weakest papers lose most values to supplementary data; median paper: 91%

6. Model error types

Curator classification of 315 annotated model errors, run of 28 Jul 2026



Missed: data in supplementary	38% (119)	Incomplete list	7% (23)
Wrong value	23% (72)	Experiment-level error	5% (16)
Missed: data in main text	15% (46)	Data from another article	3% (9)
Value from another comparison	8% (25)	Wrong form or unit	1% (5)

Supplementary-only data: the leading error type, the hallmark of the two weakest papers and the cause of all 3 missed experiments; the main limitation of the current version.

Where we are now

Run of 28 Oct 2026

+ minor adjustments of the instructions
+ improvements in the supplementary reader

Latest Gemini for text, GPT-6 Astra for images: cost $20, no improvement

87.7% correct fields

from 86.1% on 26 Sep 2026

Conclusions & next steps

Based on the run of 26 Sep 2026

- Eligible experiments identified reliably; all experiment–control pairs follow the inclusion rule.
- Most data recovered from the main text; supplementary-only data is the main limit.
- **Error analysis: 91–92% accuracy achievable with current models.**

Before release

All low-confidence fields and a random 10% of papers re-checked; records labelled as AI-extracted, with accuracy and error analysis.

Roadmap

- **Supplementary** — improve reading of supplementary files
- **368 papers** — validate on all lifespan papers in Open Genes
- **New papers** — add search and download of new papers
- **Open Genes** — integrate the pipeline into its infrastructure

AI-extractor · Open Genes · NAR 2024

Corresponding author: kat.rafikov@gmail.com

Funding: CONICET; Open Longevity